Overview

Welcome to the Mercor Cheating Detection Competition, a Kaggle community event hosted by Mercor. Your goal in this challenge is to build a model that predicts whether a candidate is engaging in cheating behavior during an interview. The dataset includes anonymized behavioral features, platform activity signals, and a social graph that captures relationships between users across the network.

Cheating detection in real interview environments is challenging because most labeled cases come from candidates who were already flagged by internal systems or human reviewers, rather than from the broader population. To mirror these conditions, this competition provides both manually reviewed ground-truth labels and a large unlabeled sample drawn from the wider candidate pool. This structure encourages participants to address class imbalance, sampling bias, semi-supervised learning, and graph-based modeling techniques.

To be eligible for prizes, participants must also apply through the Mercor registration form linked below. This application ensures that your Kaggle account is associated with your submission and allows Mercor to contact and pay winners after the competition concludes.

https://work.mercor.com/jobs/list_AAABmuZ8XM29hAHcM8NNgLXq/mercor-cheating-detection-kaggle-contest

Description

You are tasked with predicting whether a candidate in an online talent marketplace is engaging in cheating behavior during an interview, using anonymized activity features and a social graph.

Submissions are evaluated using a cost-based metric that reflects real-world operational impact: false negatives (missing confirmed cheating) incur the highest penalty, false positives in high-confidence auto-blocking lead to candidate loss costs, manual review operations incur a smaller review cost, and correct decisions incur no cost. Your model's predicted probabilities are translated into three operational decision regions (auto-pass, manual review, auto-block), and the leaderboard ranks models by the minimum achievable total cost across optimal thresholds.

Data fields: Each candidate record includes anonymized behavioral features (feature_001 through feature_018), a high_conf_clean flag identifying high-confidence not cheating unlabeled examples, and is_cheating labels for manually reviewed cases.

Train/test setup: The training set contains manually reviewed labeled candidates as well as a large set of unlabeled high-confidence examples. Test and holdout sets include only labeled candidates for evaluation. A complete social graph is also provided for graph-based detection and relational modeling.

Submission format: Predict a cheating probability (0.0 to 1.0) for each candidate in the test set. The evaluation function will automatically search for the optimal decision thresholds that minimize total operational cost.

Evaluation

Submissions are evaluated using a **cost-based metric** that reflects real-world operational costs in cheating detection. The evaluation function automatically searches for optimal decision thresholds that minimize total cost across three decision regions:

Decision Regions

- **Auto-pass** (low cheating risk): Candidate proceeds without review
- **Manual review** (medium risk): Candidate is flagged for human review
- **Auto-block** (high cheating risk): Candidate is removed from consideration without review

Cost Structure

- **False Negative** (cheating passes through): $600
- **False Positive in auto-block region**: $300
- **False Positive in manual review region**: $150

- **True Positive requiring manual review**: $5
- **Correct auto-pass or auto-block**: $0

Your score on the leaderboard is the negative of the minimum total cost found across all threshold combinations (higher scores are better). This metric rewards models that make confident, correct decisions while penalizing costly operational errors.

Submission File

For each `user_hash` in the test set, you must predict a probability that the candidate is cheating. The file should contain a header and follow this format:

```
user_hash,prediction
a3f2e1b92c8d4567,0.05
b8d4c2a71f9e3a82,0.87
c9e5d3f12a4b7c89,0.42
etc.
```

Prizes

Kaggle Prizes Available: $5,000

- First Prize: $2,500
- Second Prize: $1,400
- Third Prize: $700
- Fourth Prize: $300
- Fifth Prize: $100

These prizes are open to everyone in the Kaggle community.

Note that we will have a separate **identical prize pool** with the same structure reserved for students of undergraduate and graduate economic institutions in North America. **Campus Prizes Available: $5,000**

- First Prize: $2,500
- Second Prize: $1,400
- Third Prize: $700
- Fourth Prize: $300
- Fifth Prize: $100

Notes:

- It is possible to win awards in both tracks.
- Eligibility for the campus prize will be determined via proof of student status during form submission.
- For any questions, please leave a note in the discussion!

Citation

Drew Kringel, Peter Zhang, and Austin Robert Mann. Mercor Cheating Detection. https://kaggle.com/competitions/mercor-cheating-detection, 2025. Kaggle.